



ORACLE

Cloud Operations for ExaDB-D

Backup & Recovery Operations Scenarios

Operations Advisory Team

Office of CTO | EMEA Solution Engineering

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Agenda

- Introduction
- Backup & Recovery options
 - Oracle Managed Backups
 - User Configured Backups
 - Custom RMAN Backups
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Introduction

Introduction



This Workshop provides information about Exadata Database Service on Dedicated Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VMC, OH, DBs, DG, scaling and patching

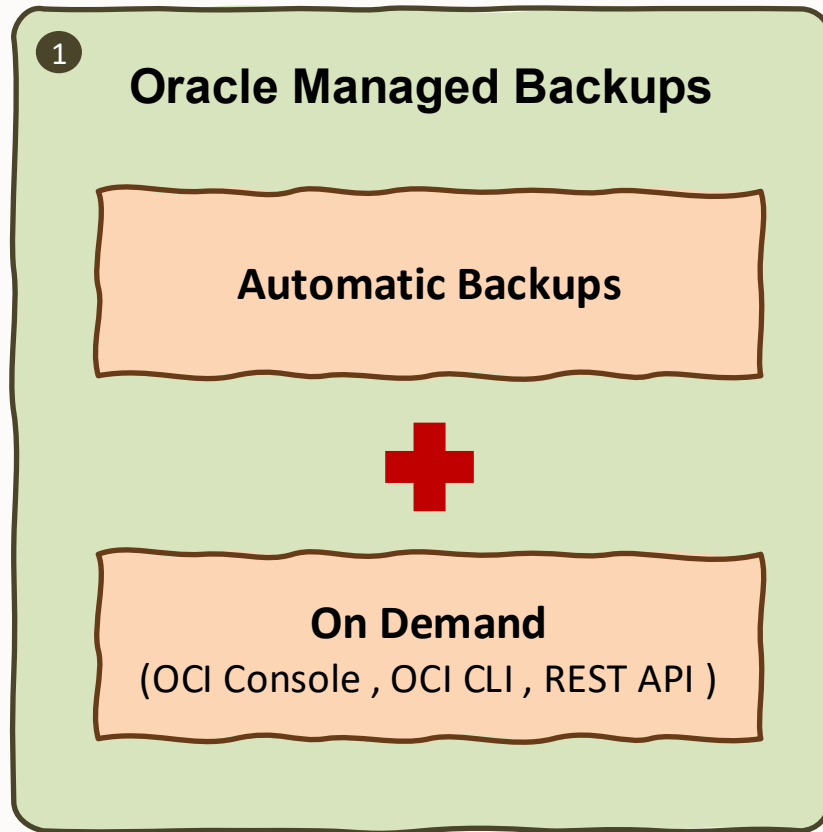
The code in this Workshop is just sample code and can be incomplete.

To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

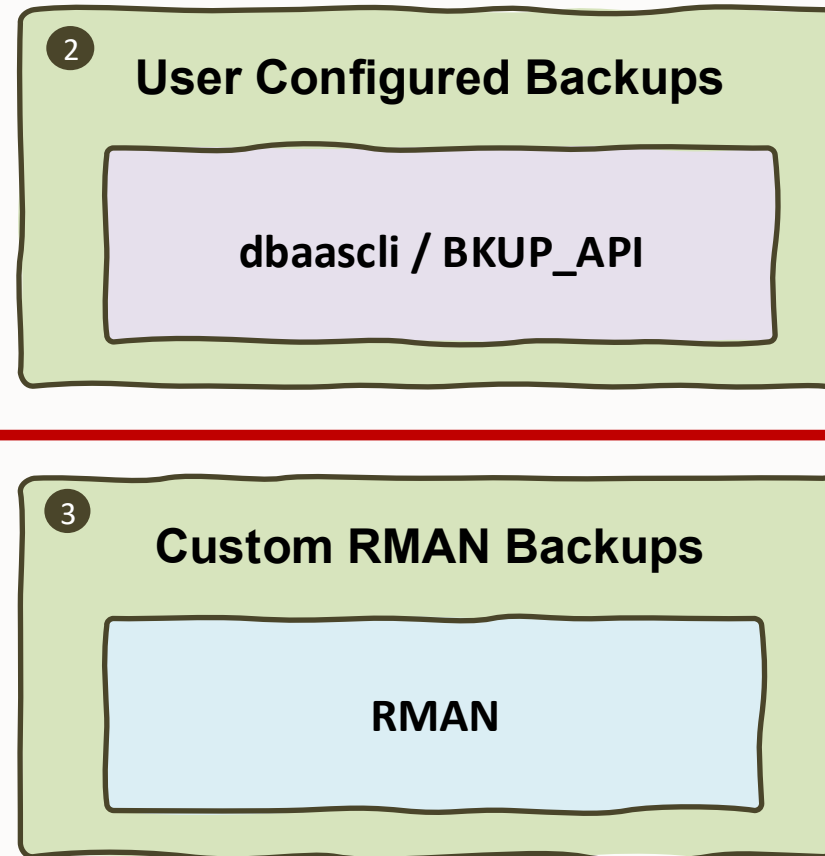
Backup & Recovery options

Backup Options

Managed by OCI Control Plane



Managed from the Database Host



Oracle Managed Backups

Oracle Managed Backups

Oracle's recommended backup solution for databases running on Exadata Database Service on Dedicated Infrastructure.

Management Model

- Managed through the OCI Control Plane.
- Integrated with OCI Console, OCI CLI and REST APIs.

Key Advantages

- Simplified administration.
- Minimal operational effort.
- Fully managed backup lifecycle.
- Synchronized with the OCI Control Plane.
- No RMAN scripting required.

Backup Types

- Automatic scheduled backups.
- On-Demand backups.

Recovery Capabilities

- Point-in-Time Recovery (PITR).
- SCN Recovery.
- Timestamp Recovery.
- Latest Backup Recovery.

Supported Backup Destinations

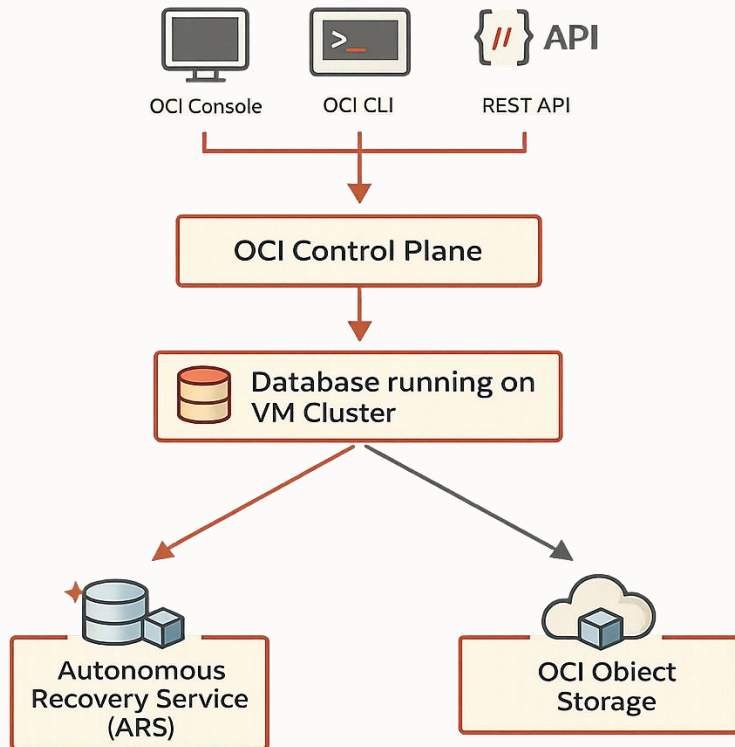
- Autonomous Recovery Service (Default option).
- OCI Object Storage*.

Best Fit

- Organizations seeking a fully managed backup solution with native OCI integration and minimal operational overhead.



Operational Considerations



Scheduling

- Automatic backups follow the configured backup policy.
- On-demand backups do not modify the backup schedule.

Operational Considerations & Limitations

- Designed for Oracle-managed workflows.
- Limited customization compared to native RMAN.
- Automatic and On-Demand backups are designed to work together.
- Oracle-Managed Backups cannot be mixed with customer-managed backup strategies.
- RMAN may be used for some restore operations (Check documentation).
- Do not use RMAN to create backups or modify backup settings while Oracle-Managed Backups are enabled.
- Disable Oracle-Managed Backups before adopting a RMAN backup strategy.
- Supported Backup Targets: CDB and PDBS.

Recovery workflow

- Recovery operations started from OCI Console, OCI CLI or REST APIs.

User Configured Backups

User Configured Backups

Customer-managed backups configured from the database host using dbaascli.

Management Model

- Managed from the database host.
- Backup metadata is not synchronized with the OCI Control Plane.

Key Advantages

- Direct access to backup destinations.
- Oracle-provided backup automation (dbaascli/BKUP_API*).
- Suitable for custom operational workflows.
- Can be used to perform destination-specific operations.

OCI Integration

- No OCI Console integration.
- No OCI API integration.
- Not visible from the OCI Control Plane.

Recovery Capabilities

- Point-in-Time Recovery (PITR).
- SCN Recovery.
- Timestamp Recovery.
- Latest Backup Recovery.

Supported Backup Destinations

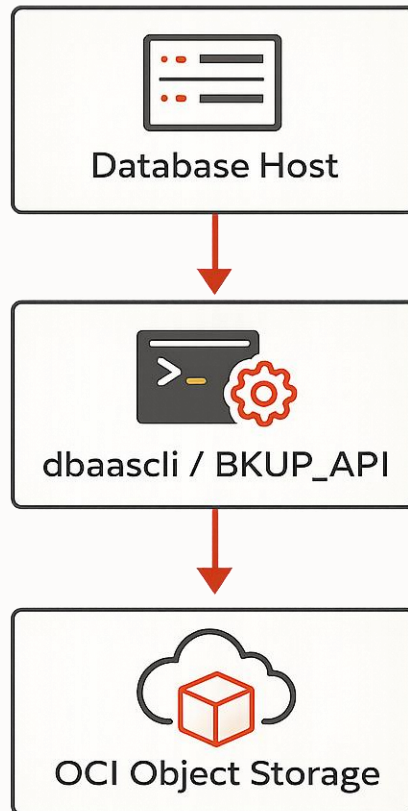
- OCI Object Storage.

Best Fit

- Organizations requiring direct access to backup destinations or custom backup workflows outside the OCI Control Plane.



Operational Considerations



Scheduling

- Backup execution and scheduling are managed by the customer.
- Backup schedules are configured using dbaascli commands.

Operational Considerations & Limitations

- Backup lifecycle is managed from the database host.
- Oracle does not recommend this approach for general deployments..
- dbaascli can coexist with Oracle-Managed Backups for supported operations.
- Configuration changes are limited to parameters compatible with Oracle Managed Backups.
- For unsupported backup configuration changes, Oracle recommends using a single backup management model.
- Supported Backup Targets: CDB, PDB and TDE Wallet.

Recovery workflow

- Recovery operations are performed using dbaascli database recover.
- RMAN may be used for supported recovery operations.

dbaascli Command Reference

Configure a Database backup

```
$ dbaascli database backup --dbName <value> --configure { --configFile <value> | --enableRTRT | --disableRTRT  
| --disableCatalog | --deleteImmutableConfiguration } 1...
```

Initiate a Database backup

```
$ dbaascli database backup --dbName <value> --start {--untilTime <value> [--nonUTC]  
| --untilSCN <value> | --latest } 1...
```

Initiate a Database recover

```
$ dbaascli database recover --dbName <value> --start [--level0] [--level1]  
{[--archival --tag <value>] | [--archivelog]} 1...
```

Initiate a PDB backup

```
$ dbaascli pdb backup --dbname <value> --pdbName <value> --start {[--level1] | [--archival --tag <value>]} 1...
```

Initiate a PDB recover

```
$ dbaascli pdb recover --dbname <value> --pdbName <value> --start {--untilTime <value> [--nonUTC]  
| --untilSCN <value> | --latest } 1...
```

Backup TDE wallets to local disk

```
$ dbaascli tde backup {[--dbname <value>] | [--allDBs]} [--backupLocation <value>] [--executePrereqs] 1...
```

Custom RMAN Backups

Custom RMAN Backups

Customer-managed backup strategy using native RMAN scripts.

Management Model

- Fully managed by the customer (Native RMAN scripts).
- Backup metadata is not synchronized with the OCI Control Plane.

Key Advantages

- Full backup strategy customization.
- Reuse existing RMAN backup and recovery scripts.
- Customer-defined backup destinations.
- Complete control over backup scheduling and retention.

OCI Integration

- No OCI Console integration.
- No OCI API integration.
- Not visible from the OCI Control Plane.

Recovery Capabilities

- Point-in-Time Recovery (PITR).
- SCN Recovery.
- Timestamp Recovery.
- Latest Backup Recovery.

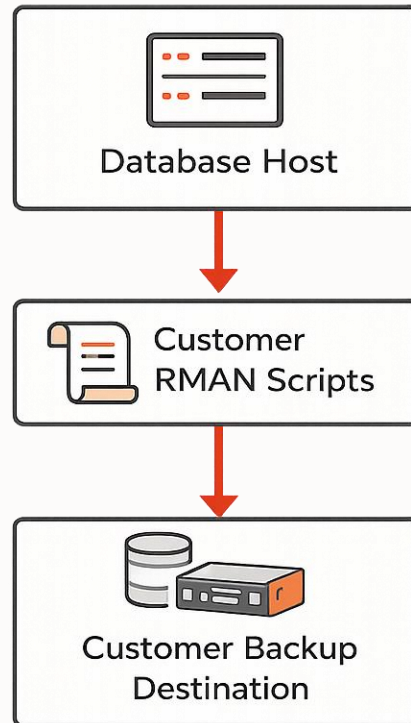
Supported Backup Destinations

- Customer-defined backup destinations.

Best Fit

- Organizations with established RMAN frameworks requiring complete control over backup and recovery operations.

Operational Considerations



Scheduling

- Backup scheduling is fully customer-defined.
- Backup execution is controlled by customer RMAN scripts or external schedulers.

Operational Considerations & Limitations

- Full ownership of the backup and recovery strategy.
- Existing RMAN backup and recovery scripts can be reused.
- Oracle-Managed and User-Configured backup strategies should not be used concurrently with Custom RMAN backups.
- Disable Oracle-Managed Backups before adopting a manual RMAN backup strategy.
- Customer is responsible for TDE Wallet backups.
- Customer manages backup retention, lifecycle and monitoring.
- Supported Backup Targets: CDB and PDB.

Recovery workflow

- Recovery operations are performed using native RMAN commands.
- Recovery procedures are fully customer-managed.

Summary

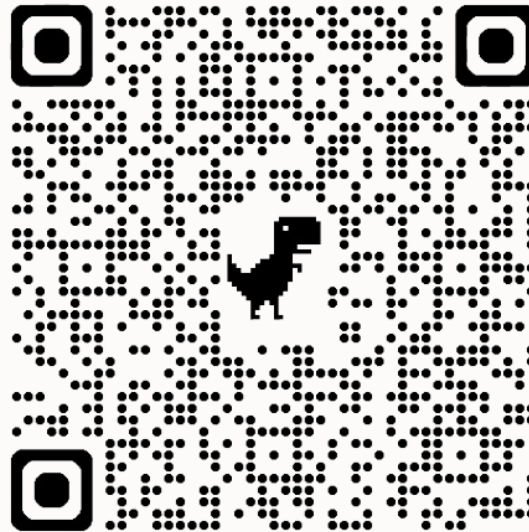
Summary



Backup type	Oracle Managed	User Configured	Native RMAN
Management Model	OCI Control Plane	Local VM Command	RMAN commands
OCI Control Plane Integration	Yes	No	No
Backup Scheduling	Automatic + On-Demand	Customer-defined	Customer-defined
Backup Destination	ARS / OCI Object Storage	OCI Object Storage	Customer-defined
Recovery Operations	PITR / SCN / Timestamp	PITR / SCN / Timestamp	PITR / SCN / Timestamp
Backup Scope	CDB / PDB	CDB / PDB / TDE Wallet	CDB / PDB
TDE Wallet	Managed by service	dbaascli	Customer-managed
Operational Effort	Low	Medium	High
Oracle Recommendation	★		



Stay tuned



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GitHub Repository

Thank you

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